Two New Private Spacecraft Launch to the Moon

Two private spacecraft are on their way to the moon to carry out separate missions.

The landers launched January 15 from the American government's Kennedy Space Center in Florida. The private company SpaceX used its Falcon 9 launch vehicle to fly the landers into space. The two separated from Falcon 9 about one hour into the flight.

Mission leaders said the launch went exactly as planned, with no problems reported. The trip to the moon will take some time. One spacecraft is expected to land on the lunar surface in early March, while the other should touch down in late May or early June.

In February 2024, the first private spacecraft completed the first U.S. moon landing in more than 50 years. The lander, called Odysseus, was developed by the Texas-based company Intuitive Machines. The spacecraft experienced some technical problems but was able to carry out several science experiments before powering down permanently on the moon in late March.

This photo provided by Firefly Aerospace in January 2025 shows the Blue Ghost lunar lander in a clean room. (Firefly Aerospace via AP)

For this current mission, Texas-based Firefly Aerospace developed Blue Ghost, one of the two landers launched. The other, named Resilience, belongs to Japanese company ispace. Both are designed to collect data and materials to support several planned moon missions – some including astronauts – in coming years.

The Blue Ghost lander is targeting a landing site near a volcanic structure called Mons Latreille. It is a 480-kilometer basin that sits in the northeast quarter of the near side of the moon.

The American space agency NASA says the 2-meter-tall Blue Ghost is carrying 10 NASA science and technology instruments. They aim to "gather valuable scientific data studying Earth's nearest neighbor," the agency said.

NASA's Artemis program aims to send astronauts to the moon for the first time since the Apollo 17 mission in 1972. The next planned flight in that program is Artemis II, which is set to launch in April 2026. In that mission, four astronauts will fly NASA's Orion spacecraft more than 400,000 kilometers on a trip around the moon.

This photo provided by SpaceX in January 2025 shows the Firefly Aerospace Blue Ghost lunar lander being placed in a rocket fairing. (SpaceX via AP)

Nicola Fox is the associate administrator for NASA's Science Mission Directorate in Washington D.C. She said in a statement the agency's cooperation with private companies is "a critical part of bringing humanity back to the moon."

Fox added that NASA chose the new experiments partly because of information learned from NASA's Apollo space program, which began in the 1960s. She said the current mission seeks to ensure "the safety and health of our future science instruments, spacecraft, and, most importantly, our astronauts on the lunar surface."

Blue Ghost's equipment includes a tool to collect dirt and another to dig a hole for measuring temperatures below the lunar surface. The spacecraft is also carrying a device built to measure light reflections to be used with lasers to better measure the distance between Earth and the moon.

In addition, Blue Ghost is carrying instruments to examine the structure and density of areas beneath the lunar surface. Other equipment will seek to capture X-ray images of the edge of Earth's magnetic field.

This photo provided by ispace in January 2025 shows the company's Micro Rover. (ispace via AP)

The ispace lander Resilience is carrying an exploring vehicle, called a rover, to the moon. The five-kilogram rover is designed to collect lunar soil and other materials from the surface.

Resilience is also carrying equipment and instruments to complete several experiments for Japanese companies and other organizations. One of the experiments will test an electrolysis device designed to separate water into hydrogen and oxygen. Such a device could help future astronauts better use water resources on the moon and produce rocket fuel.

Other experiments set for the Resilience mission include food production tests and the deployment of a "deep space radiation probe." The instrument is designed to collect detailed measurements of ionizing radiation in space.

This photo provided by ispace in January 2025 shows the Resilience lunar lander. (ispace via AP)

NASA has said it is paying $101 million to Firefly for the mission and another $44 million for the experiments. Officials from ispace did not report how much its mission would cost.

It is the second moon mission for ispace. During the last one, Japan's space agency JAXA successfully launched its SLIM spacecraft to the moon in January 2024. But the lander touched down imperfectly, causing some communication and power problems.

However, Japanese space officials reported they had stayed in communication with SLIM through late April. During this time, they said the spacecraft was able to collect valuable data about the landing and surrounding area.

I'm Bryan Lynn.

Bryan Lynn wrote this story for VOA Learning English, based on reports from NASA, The Associated Press and Agence France-Press

Quiz - Two New Private Spacecraft Launch to the Moon

Vocabulary

mission - n. ภารกิจ (ในการสำรวจอวกาศ) การเดินทางของยานอวกาศไปยังเป้าหมายพร้อมกับภารกิจที่คาดว่าจะดำเนินการ

lunar - adj. เกี่ยวกับดวงจันทร์ คล้ายดวงจันทร์ หรือมาจากดวงจันทร์

reflect - v. สะท้อน (ถ้าพื้นผิวสะท้อนความร้อน แสง ฯลฯ หมายถึงการส่งแสง ฯลฯ กลับไปและไม่ดูดซับไว้)

electrolysis - n. กระบวนการอิเล็กโทรไลซิส การใช้กระแสไฟฟ้าเพื่อทำให้เกิดการเปลี่ยนแปลงทางเคมีในของเหลว

ionize - v. ทำให้เป็นไอออน (การทำให้อะตอมหรือกลุ่มอะตอมขนาดเล็กมีประจุไฟฟ้าโดยการเพิ่มหรือสูญเสียอิเล็กตรอนอย่างน้อยหนึ่งตัว)

Reading Comprehension Quiz

1. What is the main subject of this article based on the opening statement in paragraph 2?

A) Government-funded space exploration initiatives.

B) Two private spacecraft currently traveling to the moon.

C) Upcoming missions involving human astronauts.

D) Historical data on past moon landings.

2. According to paragraph 3, from which location did the landers launch?

A) A private facility operated by SpaceX.

B) The Kennedy Space Center in Florida.

C) A spaceport located in Japan.

D) The headquarters of Firefly Aerospace in Texas.

3. Which company utilized its Falcon 9 launch vehicle to send the landers into space, as mentioned in paragraph 3?

A) Intuitive Machines

B) Firefly Aerospace

C) SpaceX

D) ispace

4. How did mission leaders describe the initial launch of the spacecraft, according to paragraph 4?

A) It experienced minor delays but was ultimately successful.

B) It went exactly as planned with no reported issues.

C) It required immediate adjustments after liftoff.

D) It faced unexpected technical challenges at launch.

5. Based on paragraph 4, when are the two spacecraft expected to land on the lunar surface?

A) Both spacecraft are scheduled for early March.

B) Both spacecraft are scheduled for late May or early June.

C) One in early March and the other in late May or early June.

D) The article does not specify landing dates.

6. What was the name of the first private U.S. spacecraft to successfully land on the moon in over 50 years, as detailed in paragraph 5?

A) Blue Ghost

B) Resilience

C) Falcon 9

D) Odysseus

7. According to paragraph 5, what did the Odysseus lander manage to do before powering down permanently, despite facing technical problems?

A) It returned to Earth safely with lunar samples.

B) It successfully deployed a small exploration rover.

C) It carried out several science experiments.

D) It assisted in the launch of the current mission.

8. Which company was responsible for developing the Blue Ghost lander, one of the two landers launched in the current mission, as stated in paragraph 7?

A) Intuitive Machines

B) SpaceX

C) Firefly Aerospace

D) ispace

9. What is the shared objective of both the Blue Ghost and Resilience landers in this mission, according to paragraph 7?

A) To establish a permanent human habitat on the moon.

B) To test new rocket propulsion technologies.

C) To collect data and materials to support future moon missions.

D) To conduct speed trials for lunar landing.

10. Where is the Blue Ghost lander specifically targeting its landing on the moon, according to paragraph 8?

A) Near a previously used U.S. landing zone.

B) In the far side's southern hemisphere.

C) Near a volcanic structure named Mons Latreille.

D) At one of the moon's polar regions.

11. How many NASA science and technology instruments is the 2-meter-tall Blue Ghost lander carrying, as mentioned in paragraph 9?

A) 2

B) 5

C) 10

D) 480

12. What is the primary goal of NASA's Artemis program, as described in paragraph 10?

A) To establish commercial space travel for tourists.

B) To send astronauts to the moon for the first time since 1972.

C) To explore the possibility of human life on Mars.

D) To develop advanced robotic exploration tools.

13. According to Nicola Fox, associate administrator for NASA's Science Mission Directorate, what is considered 'a critical part of bringing humanity back to the moon'?

A) Increasing the budget for independent research.

B) Enhancing international space agency collaborations.

C) NASA's cooperation with private companies.

D) Focusing solely on robotic missions.

14. Based on paragraph 13, what is one of the crucial aspects the current mission aims to ensure, as highlighted by Nicola Fox?

A) The rapid deployment of commercial satellites.

B) The safety and health of future astronauts on the lunar surface.

C) The development of advanced artificial intelligence for spacecraft.

D) The discovery of new extraterrestrial life forms.

15. Which of the following is NOT described as an instrument carried by the Blue Ghost lander in paragraph 14?

A) A tool designed to collect dirt from the lunar surface.

B) A device built to measure temperatures below the lunar surface.

C) Equipment designed to separate water into hydrogen and oxygen.

D) A device to measure light reflections for calculating Earth-moon distance.